

MOOSE 1 NNE, WY, USA

WMO: 725740

Lat: 43.6615N

Lon: 110.7120W

Elev: 1971

StdP: 79.78

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 04131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.9	(c) -24.1	(d) -29.5	(e) 0.3	(f) -26.6	(g) -26.7	(h) 0.4	(i) -24.1	(j) 4.9	(k) -2.3	(l) 4.6	(m) -2.5	(n) 1.0	(o) N/A	(p) 0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	29.1	12.6	27.6	12.4	26.0	12.0	14.7	23.8	13.9	23.2	13.1	22.7	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	11.0	16.5	10.6	10.1	16.0	9.5	9.4	15.6	47.9	23.7	45.4	23.1	43.3	22.6	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.8	3.2	-32.4	31.2	2.8	0.9	-34.3	31.8	-35.9	32.3	-37.5	32.8	-39.5	33.4		
			-32.1	16.7	3.2	0.5	-34.4	17.0	-36.3	17.3	-38.1	17.6	-40.5	17.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-10.5	-7.8	-3.0	2.0	7.4	12.0	16.5	15.1	10.8	3.9	-2.8	-9.5
	DBStd	10.30	6.39	5.86	4.90	3.93	3.48	3.35	2.38	2.46	3.43	4.12	5.49	6.65
	HDD10.0	3107	637	499	403	241	95	19	0	2	31	191	384	605
	HDD18.3	5647	895	732	662	491	341	190	66	101	226	446	634	863
	CDD10.0	513	0	0	0	1	13	79	201	161	56	3	0	0
	CDD18.3	10	0	0	0	0	0	1	8	2	0	0	0	0
	CDH23.3	1067	0	0	0	0	4	126	539	336	61	1	0	0
	CDH26.7	221	0	0	0	0	0	21	136	59	6	0	0	0
(14) Wind	WSAvg	1.5	1.4	1.8	1.7	1.8	1.6	1.4	1.5	1.2	1.3	1.4	1.4	1.4
(15) Precipitation	PrecAvg	560	67	49	47	42	52	49	27	33	36	43	53	72
	PrecMax	766	117	124	110	64	100	109	81	59	76	80	119	193
	PrecMin	390	14	18	19	12	9	15	6	8	8	5	11	29
	PrecStd	111	31	21	23	14	25	28	20	16	16	20	25	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.7	11.3	19.3	23.8	29.1	31.2	30.0	27.3	20.9	12.9	5.2
		MCWB	0.1	1.2	4.0	7.3	9.9	12.8	13.3	12.5	12.4	8.9	5.2	0.4
	2%	DB	1.7	3.5	7.9	15.3	21.6	26.6	29.7	28.3	25.1	17.5	10.1	3.3
		MCWB	-0.6	0.4	2.6	5.8	9.3	12.2	13.1	11.9	11.0	7.6	3.9	0.7
	5%	DB	0.4	2.3	6.1	12.5	19.2	24.6	28.3	26.9	23.4	15.0	7.5	1.6
		MCWB	-1.3	-0.2	1.6	4.7	8.3	11.7	12.8	11.7	10.0	6.6	2.4	-0.4
	10%	DB	-1.0	0.9	4.3	9.7	16.5	22.5	26.7	25.3	21.1	12.2	5.2	0.2
		MCWB	-2.3	-0.9	0.8	3.4	7.6	10.9	12.6	11.5	9.5	5.6	1.2	-1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.9	4.3	8.5	11.1	14.8	16.1	15.2	14.6	10.4	5.8	2.0
		MCDB	2.5	4.4	9.9	18.0	20.3	25.8	24.9	23.3	21.6	18.4	11.3	3.8
	2%	WB	0.2	1.2	3.1	6.5	9.8	13.3	15.0	14.1	12.7	8.9	4.2	1.0
		MCDB	1.5	3.4	7.2	15.1	18.7	23.9	24.2	22.6	20.3	15.7	9.5	2.5
	5%	WB	-0.6	0.4	2.2	5.0	8.8	12.3	14.2	13.3	11.6	7.5	2.8	0.0
		MCDB	0.7	2.4	5.4	11.1	17.2	22.6	23.7	22.3	20.2	14.2	6.5	1.4
	10%	WB	-2.1	-0.3	1.4	3.7	7.8	11.4	13.6	12.5	10.6	6.2	1.5	-1.3
		MCDB	-0.6	1.5	4.2	8.9	15.3	21.2	23.3	22.0	19.0	11.6	4.3	0.6
(35) Mean Daily Temperature Range	MDBR		10.8	10.7	11.4	11.9	14.3	16.9	19.5	19.5	17.8	13.9	11.2	10.3
	5% DB	MCDBR	9.5	8.9	13.0	17.9	20.3	21.8	22.2	22.6	22.6	19.5	14.5	8.0
		MCWBR	8.0	6.0	7.6	9.3	9.1	9.3	8.8	8.9	9.7	10.3	9.0	6.2
	5% WB	MCDBR	8.4	6.9	10.7	16.2	17.5	19.9	19.3	18.4	17.9	16.7	13.5	7.3
		MCWBR	7.5	5.3	6.5	8.5	8.2	8.8	8.2	7.9	8.0	8.9	8.4	5.9
(40) Clear-Sky Solar Irradiance	taub		0.234	0.245	0.261	0.298	0.304	0.311	0.325	0.333	0.301	0.280	0.250	0.232
	taud		2.354	2.313	2.330	2.358	2.489	2.518	2.502	2.461	2.553	2.572	2.516	2.403
	Ebn at Noon		920	965	988	968	966	956	938	922	934	917	900	890
	Edh at Noon		77	96	109	116	105	103	104	104	88	75	66	66
(44) All-Sky Solar Radiation	RadAvg		1.84	2.74	4.00	5.19	5.84	6.93	7.26	6.13	4.79	3.03	1.87	1.39
	RadStd		0.18	0.33	0.35	0.43	0.61	0.62	0.38	0.36	0.43	0.27	0.20	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page